

Multiple Choice

1. **Answer:** B

Options: A) True, False; B) True, True; C) False, False; D) False, True

Note: Correct option: B - True, True

2. **Answer:** A

Options: A) All trapezoids are rectangles because they have at least one pair of parallel sides.; B) All pentagons are heptagons because they have more than 4 sides.; C) All hexagons are triangles because they have at least 3 sides.; D) All triangles are polygons because they have less than 4 sides.; E) All rectangles are squares because they have 4 right angles.

Note: Correct option: A - All trapezoids are rectangles because they have at least one pair of parallel sides.

3. **Answer:** C

Options: A) 10; B) 30; C) 23; D) 15; E) 18

Note: Correct option: C - 23

4. **Answer:** C

Options: A) 2200; B) 1800; C) 2041; D) 2100; E) 1700

Note: Correct option: C - 2041

5. **Answer:** D

Options: A) 1:03; B) 60:15; C) 4:01; D) 1:04; E) 3:01

Note: Correct option: D - 1:04

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '8.984'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.186'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
Options: A) True; B) False
Note: Actual answer is '386.4158898'.

Long Form Response

11. **Answer:** QED $\&= (5+2i)(i)(5-2i)$
 $\&=i(25-(2i)^2)$
 $\&=i(25+4)$
 $\&=29i$.
Note: Students should show all steps in their mathematical solution.
12. **Answer:** $-\frac{49}{12}$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.